

Fahrkartenkontrolle: A Verification-First Harness for Reliable Tool-Using Agents

Anton Wolter
Aarhus University

Abstract

Deployment of tool-using LLM agents is gated less by whether they can solve a task than by whether they solve it *every* time. We describe *Fahrkartenkontrolle*, our CAR-bench Challenge entry: a single open-weight model wrapped in six policy-preserving grounding-and-defaults rules and one self-check pass that blocks claims of actions no successful tool call supports. On the full public test split the harness lifts Pass³ from 52.0% to 71.3%, almost entirely by suppressing fabricated capabilities. We report with equal weight what did not help: prompt refinements trade one task category against another where the verification pass composes; self-consistency voting triples token cost with no gain, because the model’s residual errors are systematic rather than random; and a controlled language ablation places the German-instruction advantage that motivated our system name within replication noise. A post-submission screen in which a second model family runs the self-check gave our best screening score; the gap that remains between occasional and consistent success is exactly the failure mode the benchmark is designed to expose.

1 Introduction

Benchmarks for tool-using LLM agents have historically asked whether the agent can solve a task at all; for a user-facing assistant the operative question is whether it solves the task *every* time it is asked. CAR-bench [Kirmayr *et al.*, 2026] makes this distinction the core of its design: an in-car voice assistant with 58 interconnected tools, 19 domain policies, mutable vehicle state, and an LLM-simulated driver, scored by Pass^k (the fraction of tasks solved in *all* of *k* independent trials), with dedicated *hallucination* tasks (a capability does not exist and must be refused) and *disambiguation* tasks (ambiguity must be resolved from stored preferences and policy defaults, not by interrogating the user).

This paper describes our entry to the IJCAI-ECAI 2026 CAR-bench Challenge (Track 1). The system is named *Fahrkartenkontrolle* (German for ticket inspection) after its central mechanism: before any user-facing reply leaves the agent, every claim of a performed action must present its ticket, a successful tool call in the conversation log. The final

configuration is deliberately small: around one open-weight model (GLM, temperature 0), a prompt wrap appends six grounding-and-defaults rules to the evaluator’s policy text, and a single self-check pass rewrites unsupported claims. On the full public test split this lifts Pass³ from 52.0% to 71.3% (+19.3 points), with essentially the whole gain coming from hallucination suppression; several further mechanisms we built did not survive measurement (Sec. 4). The campaign behind this configuration ran more than thirty evaluation rounds with a run-cited insights ledger,¹ and we believe its lessons are more transferable than the score:

- **Prompting trades; verification composes.** Every prompt refinement that improved one task category paid for it in another; the self-check improved a category without giving one back, because it rewrites unsupported *text* and never suppresses an *action*.
- **Sampling cannot fix systematic errors.** Self-consistency voting tripled token spend without improving Pass³: independent samples vote for the same wrong action.
- **Instruction language deserves controlled tests, not folklore.** German-language rules appeared to induce behavior that semantically equivalent English rules did not; a replicated controlled ablation places the differences within replication noise (Sec. 4).
- **Measurement is the hard part.** A 45-task screening subset misranked configurations, the user-simulator choice dominated early results, and replication spreads reached 4.4 points on the champion and 8.9 on a language arm (Sec. 4).

2 Related Work

AgentBench [Liu *et al.*, 2024] and its successors evaluate agent capability on single attempts; τ -bench [Yao *et al.*, 2025] introduced the Pass^k metric and the tool-agent-user triad that CAR-bench [Kirmayr *et al.*, 2026] extends with hallucination and disambiguation tasks. Our results reproduce the family’s central claim: champion and baseline differ by 2.6 points in Pass@3 but 19.3 points in Pass³: the models rarely lack capability, they lack consistency. Our self-check

¹Agent code, scenario configs, the run registry, the insights ledger, and the generated results dashboard (experiments/dashboard.html): <https://github.com/multi-tasking/Car-Bench>

pass is a deliberately narrow relative of self-correction methods [Shinn *et al.*, 2023; Madaan *et al.*, 2023; Dhuliawala *et al.*, 2024]: a single non-iterative pass over exactly two error classes, grounded in the deterministic tool-call transcript rather than free-form self-critique. Self-consistency voting [Wang *et al.*, 2023] improves single-shot reasoning; we report that it fails for multi-turn tool use (Sec. 4). Multilingual studies show performance varies with the *reasoning* language [Shi *et al.*, 2023; Etxaniz *et al.*, 2024], and even semantically irrelevant non-English material in an otherwise English prompt can measurably improve performance [Tu *et al.*, 2025]; we probe the complementary variable, the language of the operating instructions with rule semantics held fixed, and find its effect within replication noise (Sec. 4).

3 System

The challenge fixes a strict boundary (Fig. 1): the evaluator owns the simulated user, the tool environment, and scoring, and exchanges A2A JSON-RPC messages² with a dockerized agent under test that executes nothing itself. The agent is a linear pipeline of stages, each switchable by environment variable, so every mechanism remains in the shipped image and every claim of the form “stage *X* contributes *Y*” is testable by flipping one flag. The champion enables exactly two stages—the `v4_german` prompt wrap (stage 0) and the self-check pass (stage 5)—around a single open-weight GLM model queried at temperature 0 through a LiteLLM proxy; the rest ship disabled, with measurements in Table 1.

Stage 0: policy-preserving prompt wrap. As policy compliance is scored, the evaluator’s policy prompt is never edited but the agent may append to it. The champion appends six rules, in priority order: (1) ground every claim in tool results, and state plainly when a capability is unavailable; (2) treat every request minimally and never claim another component moved along unless a tool result says so; (3) resolve ambiguity yourself first, from stored user preferences and policy defaults, asking only as a last resort and converging on exactly one answer; (4) give unspecified parameters their policy default; (5) act immediately on confirmation; (6) policies take precedence, answer in English, keep replies short. Each rule traces to a judge-identified failure cluster from an earlier round. The rules are written in German, motivated by two untested hypotheses: that German’s morphologically explicit imperatives might read as higher-precision instructions, and that the automotive domain’s pretraining corpus is unusually rich in German-language primary text. Section 4 tests the choice directly.

Stage 5: the self-check pass (*Fahrkartenkontrolle*). Whenever a turn would end in user-facing text with no tool calls, a second LLM call inspects the draft. It receives the ordered tool-call transcript with success statuses, the recent user messages, and a deterministic pre-scan flagging components the draft mentions without a supporting successful call, and reports exactly two error classes: (i) the draft claims an action happened or will happen automatically with no supporting call, and (ii) the user confirmed an action, no call executed it, and the draft pretends it is done. Anything else is out of scope; on a violation it returns a minimally corrected reply, and on any checker failure the agent sends the original

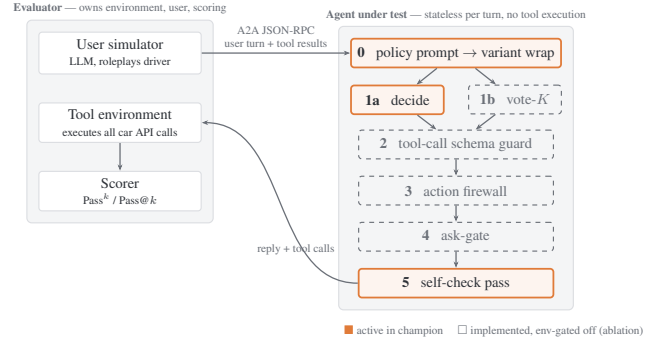


Figure 1: Runtime architecture. The evaluator owns the simulated user, the tool environment and scoring; the agent under test only decides. Every pipeline stage is independently switchable by environment variable, so each is a one-flag ablation. Stages 1a and 1b are *alternative* decision paths (a single GLM pass at $T=0$, or a K -sample vote), never chained. The champion configuration enables stage 0 (`v4_german` prompt wrap, appended to the evaluator’s policy text), path 1a, and stage 5 (self-check); 1b and stages 2–4 ship disabled with their measured effect reported in Table 1.

draft. The narrow mandate is what lets the stage compose with stage 0: it can rewrite unsupported *text* but can never suppress an *action*. The gated-off stages (self-consistency vote- K , a tool-call schema guard, an advisory action firewall, and an ask-gate that nudges one regeneration before unnecessary clarification questions) are evaluated in Sec. 4.

The agent builder. The system that produced the champion is larger than the agent itself: around the self-hosted models sits a closed experiment loop driven by a coding agent in a long-lived, largely autonomous session: it launches evaluation runs, appends structured results to an append-only registry, regenerates a dashboard, clusters failed trajectories with an LLM judge, and turns the clusters into prompt-edit proposals for the next variant, with all LLM traffic traced through the LiteLLM proxy via Langfuse and long runs executing unattended on self-hosted CI runners. In other words, we did not so much build an agent as operate an *agent builder*, and most of the code, experiments, and failure analyses behind this paper are its output. Human intervention concentrated at a few ledger-recorded decision points (validating and swapping the user-simulator model, declaring the prompt plateau, selecting the champion on tie-breaks), and which of those genuinely required a human is an open question; what we can say is that the unit under development was the builder loop, not the agent, which suggests comparing *builders* by the measured reliability of their produced agents.

4 Evaluation

Setup. All configurations use the same GLM model at temperature 0 for both the agent and (separately) the simulated user, with policy verdicts from a second model family. Temperature 0 does not make trials deterministic; serving-side nondeterminism and the stochastic simulated user remain, which is the variability Pass^k exposes: a task passes Pass^k if all k trials succeed, $\text{Pass}@k$ if at least one does. Configuration screening ran on an equal-width *train* subset of 45 tasks (15 per category) $\times$ 3 trials, always anchored against a baseline run of identical width; headline comparisons ran on

²<https://a2a-protocol.org>

the full public *test* split of 125 tasks (50 base, 50 hallucination, 25 disambiguation) $\times$ 3 trials.

Headline result. Figure 2 gives the full-test comparison: the raw model reaches 52.0% Pass³, the champion (v4_german wrap + self-check) 71.3%, at roughly +30% token cost. Essentially the entire gain is hallucination suppression (22% $\rightarrow$ 76%; Fig. 2 gives the per-category detail). The gain is almost pure *repeatability*: Pass@3 is 86.7% for the raw model and 89.3% for the champion, only 2.6 points apart, so both can almost always solve a task once and differ mainly in doing so every time. The champion’s residual $89.3 - 71.3 = 18$ -point Pass@3–Pass³ gap is trial-to-trial inconsistency, and its full-test score sits only ~ 2 points below its train-subset score, so the train-fitted rules transfer.

Prompting trades categories; verification composes. On the screening subset, every prompt-level intervention bought its target category by paying with another: the v4_german minimalism and defaults rules lifted Disambiguation Pass³ (46.7 $\rightarrow$ 60) while dropping Base (86.7 $\rightarrow$ 80), because over-minimalism suppresses required auxiliary actions, and two further refinement rounds never escaped this see-saw. The self-check is the exception: added on top, it restored Base to 86.7 *without* surrendering the Disambiguation gain. The asymmetry is structural: a prompt rule changes the policy the model acts under on every turn, so it can suppress correct behavior as easily as incorrect behavior, while the self-check only rewrites unsupported claims in outgoing text and at worst leaves the draft unchanged.

What did not help. Self-consistency voting over three sampled candidate turns tripled token spend from 6.2M to 18.7M per screening run, while *not* improving Pass³ (73.3 $\rightarrow$ 71.1). It targeted exactly the 18-point consistency gap and failed because that gap is not sampling noise: the residual errors are systematic, so independent samples vote for the same wrong action. This is why the shipped harness is a single deterministic verification pass rather than an ensemble, and it suggests closing the remaining gap requires mechanisms with information the model does not already have. The other gated-off stages fared no better (Table 1): the ask-gate shuffled categories without lifting the total, the schema guard never fired in any measured run, and the action firewall, measured once with its precondition and provenance checks active, exactly tied the champion.

What might: cross-model verification. A post-submission measurement gives the “information the model lacks” hypothesis its first data point: keeping the champion unchanged but moving the self-check pass to a second model family produced Pass³ 77.8% at screening width, above every champion replicate (73.3/68.9/68.9), at token-cost parity, with hallucination Pass¹ reaching 100%. By our own replication rule this single run, whose margin (+4.4 over the best replicate) sits inside the spread band, and whose verifier model serves as the benchmark’s failure judge, is a lead to confirm rather than a result; but it is the only intervention we measured that moved the total *up*, and it did so where the theory predicts: an independent verifier disagrees with the actor’s systematic errors where K samples of the actor agree with them.

Is the German language load-bearing? In early 15-task screening, the German-language protocol reproducibly in-

Stage	Mechanism	Champion	Measured effect
0	v4_german prompt wrap	on	trades categories
1a	LLM decision, $T=0$	on	—
1b	self-consistency vote-3	off	-2.2 pp, $3\times$ cost
2	tool-call schema guard	off	no firings observed
3	action firewall	off	± 0 , ties champion
4	ask-gate	off	± 0 , shuffles categories
5	self-check pass	on	composes, restores Base
5'	cross-model verifier	off	+4.4 pp, single run
Full public test split (125 tasks $\times$ 3)		Pass³	Pass@3
raw model		52.0%	86.7%
prompt wrap + self-check		71.3%	89.3%

Table 1: Final ablation summary. Stage 1b is the alternative decision path to 1a, not its successor. Effects for stages 1b, 3, 4, and 5' are measured on the 45-task train subset and lie within its replication spread band (Sec. 4); the headline comparison is full-test.

duced the proactive preference lookup that the disambiguation category rewards, while a semantically equivalent English protocol did not (100% vs. 80% in the discovery round, confirmed in trajectories across four trials); that observation named the system. The confirmatory test, completed with post-submission replicates, is decisive: translating the final six rules verbatim into English and Spanish and re-running at full screening width gives German 73.3/68.9/68.9, English 66.7/75.6, and Spanish 73.3/68.9. The second English replicate exceeds every German score, the per-language means are indistinguishable (70.4/71.1/71.1), and the largest swing in the campaign is *within* a single configuration (English, 8.9 points), twice the champion’s replication spread. We therefore read instruction language as interchangeable within noise on this benchmark: rule content, not language, sets the level, and disambiguation is the weakest category in every language-arm run; the ceiling is language-independent. Whether instruction language modulates tool-use compliance at sample sizes this benchmark cannot affordably reach remains open.

Measurement lessons. Three findings about *measuring* agents shaped, and in one case overturned, our conclusions. First, the user simulator dominates before the agent does: our first screening round was invalidated by a simulator model that emitted hallucination-task error markers before the agent’s first turn, and switching simulators moved the *baseline* from 60% to 86.7% on identical tasks, a larger swing than any agent-side intervention we measured. Second, subsets can misrank: at screening width the baseline tied our best prompt variants (71.1%), and only the full test split (where the same comparison is 52.0% vs. 71.3%) exposed this as a subset artifact caused by under-represented hallucination behavior. Third, replicate the champion before believing the leaderboard: re-running the identical champion configuration produced Pass³ scores 4.4 points apart (and a language-arm replicate later swung 8.9 points, overturning a paired-comparison claim), so we adopted the rule that no configuration is declared better on evidence smaller than the replication spread of the configuration it replaces.

5 Conclusion

Fahrkartenkontrolle wraps a single open-weight model in six policy-preserving grounding rules and one narrow self-check pass, lifting Pass³ on the full public test split from 52.0% to

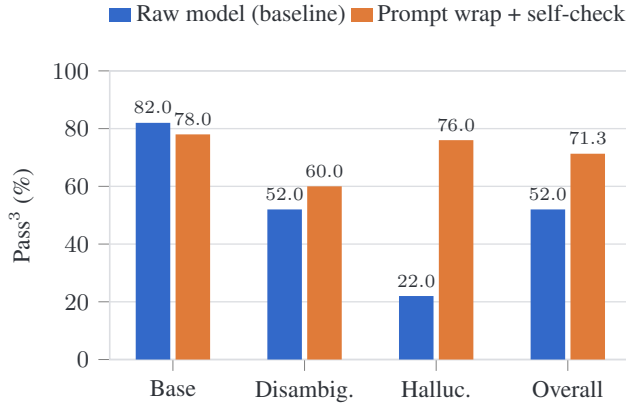


Figure 2: Headline result on the *full public test split* (125 tasks $\times$ 3 trials). The harness lifts Pass³ from 52.0% to 71.3% (+19.3 points). Essentially the entire gain is hallucination suppression (22% $\rightarrow$ 76%): the unprompted model fabricates capabilities on roughly four of five hallucination trials, the grounding rules plus self-check hold it to about one in four. Base drops slightly (82 $\rightarrow$ 78), the cost of the minimalism rule.

71.3% at roughly +30% token cost, almost entirely by suppressing fabricated capabilities. The campaign’s transferable lessons are mechanism-level: prompt rules trade task categories against each other while a bounded verification pass composes; self-consistency voting cannot close a consistency gap whose errors are systematic; and measurement discipline (simulator validation, equal-width anchoring, champion replication) decides which effects are real, including our own eponymous one, whose German-instruction advantage did not survive a controlled ablation. Limitations: all results are for one model family, with the same family simulating the user (correlated blind spots are possible); most screening ablations are single-run against within-configuration replication spreads of 4.4–8.9 points; and the rules were fitted to public train failures, with the $\sim$ 2-point train-to-test drop bounding but not eliminating overfitting.

The champion’s residual 18-point Pass@3–Pass³ gap is the benchmark’s thesis reproduced in our own system: a capable model, inconsistently right. Since sampling demonstrably does not close it, we expect progress from mechanisms that add information the model lacks (a second model family running the self-check gave our best screening Pass³ of 77.8%, single run, Sec. 4) and, because the unit we actually developed was the autonomous experiment loop of Sec. 3 rather than the agent it emitted, from comparing *agent builders* by the reliability of the agents they produce.

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AI-usage disclosure. Consistent with the agent-builder methodology of Sec. 3, large parts of the experiment harness code, the evaluation runs, and the failure analyses were

produced by an LLM-based coding agent operating under author supervision, and LLM assistance was used in drafting and editing the manuscript.

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